Statistical Learning

First Homework: Unsupervised Learning

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1. Objectives

For this homework we decided to choose the R-library open dataset NHANES (National Health and Nutrition Examination Survey). It is a dataset that shows the health data of survey takers in the year interval 2009-2010 and 2011-2012. This dataset has 10000 observations and 76 variables. We want to explore our data using different unsupervised learning tools(PCA, Factor analysis, and clustering). We aim to see how much we can reduce the dimensions of our data using PCA. This dimensionality reduction will preserve the most important variance, while enabling the identification of key patterns and relationships among health, lifestyle, and demographic variables. We will try to use factor analysis to identify any underlying relationships between the variables. Lastly, we aim to find a way to classify the individuals based on their risk of cardiovascular issues using clustering methods like k means, pam, kernel k means, and EM clustering.

2. Data processing and visualization

2.1. Variable selection

Looking at the summary of the data, we can see that there are columns that have a lot of NAs and factor variables. For our purposes, we can remove those features.

2.2. Dealing with NA values

Find which rows have the most NA values and remove the rows that have a lot of missing data. For the rest of the variables we will replace the NA values with the average of each variable.

2.3. Feature engineering

- BPRatio - abnormal BPRatio may indicate cardiac issues
- Total Bad Health Days - how many days of the year the person didn't feel good.
- SleepAdjusted shows if based on age, a person is sleeping enough.
- Cholesterol BMI Ration - the relation between cholesterol levels and body mass.
- PulseAgeRatio can indicate health problems if the ratio is abnormal for the age

2.4. Normalization - we have normalized every numerical variable that we have

2.5. Dealing with outliers

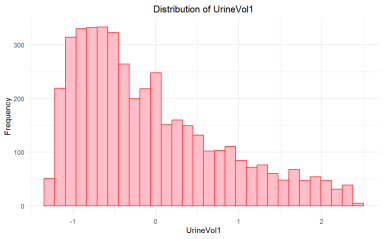
```
           Age         HHIncomeMid             Weight             Height                BMI
             0                   0                109                 88                177
         Pulse            BPSysAve            BPDiaAve           TotChol           UrineVol1
            51                 155                175                 97                174
    UrineFlow1     DaysPhysHlthBad     DaysMentHlthBad            BPRatio  TotalBadHealthDays
           374                 626                736                311                754
 SleepAdjusted CholesterolBMIRatio       PulseAgeRatio
            24                 108                395
```

From that we can see that we have many outliers in some of the columns. These features probably have high uniqueness relative to the rest of the variables, which can mess with our models, so it would be best to remove these columns. After removing the variables, which we deem unneeded, we can now remove the outliers that the remaining variables have.

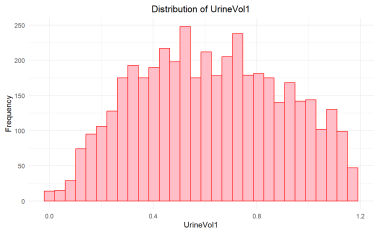
2.6. Visualization and transformation

We decided to create histograms to show the distribution of each variable. This allows us to see which variables are normally distributed, which are skewed, which are multimodal(have peaks in the histogram), or which have low variance. Overall, most variables are normally distributed, except UrineVol1 and PulseAgeRatio, which exhibit heavy right skewness. We can transform these variables to get them to be more normally distributed.

Example for UrineVol1:

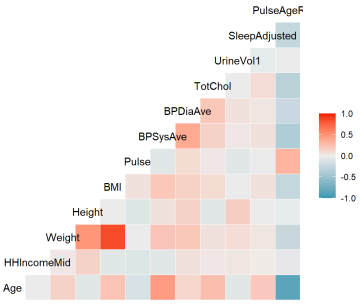


(Before)



(After)

2.7. Correlation



From this graph we can see that there aren't a lot of variables that have a high correlation between each other. Weight and Height have a strong positive relation as well as BPSysAve and BPDiaAve. Weight and BMI have a moderately strong positive relation. Variables like UrineVol1 and SleepAdjusted very weak relations with the rest. Also, PulseAgeRation shows negative relations with some of the variables.

3. PCA

The first component appears to represent a **general health index, particularly cardiovascular health and factors related to age and body weight**. This component could differentiate younger and physically healthier individuals from those at higher risk of cardiovascular health issues due to factors such as age, weight, and BMI.

The second component seems to capture a **body composition index**, focusing on characteristics that define physical development, body size, and possibly the influence of age on these metrics. This component may help differentiate individuals based on physical build, growth stage, or age-related body composition differences.

The third component seems to represent a **socioeconomic and physiological health index**. This component may differentiate individuals by socioeconomic background and how it influences certain health metrics, such as pulse, BMI, and urine output, which can all be affected by lifestyle, diet, and overall well-being.

The fourth component appears to represent a **cardiovascular and metabolic health risk index**. This component could distinguish individuals with potential risk factors for cardiovascular conditions based on their blood pressure, heart rate, cholesterol, and BMI, providing insights into overall cardiovascular and metabolic health profiles.

The fifth component seems to represent a **lifestyle, socioeconomic and cardiovascular health index**. This combination captures the interplay between lifestyle habits, economic conditions, and physiological health markers, emphasizing how these domains collectively

influence an individual's overall health profile. This component may help differentiate individuals based on how these interconnected factors impact their health outcomes.

PC6 may represent a component associated with **lifestyle and health factors influenced by socioeconomic status, sleep patterns, and physical health indicators**. This combination might differentiate individuals based on lifestyle-driven health markers or well-being factors. If interpreting this component's role, it could offer insights into how economic and behavioral factors jointly influence specific health outcomes.

PC7 appears to capture a dimension of health focused on **cardiovascular and metabolic factors**, potentially differentiating individuals based on a blend of these physiological indicators and their overall health status.

This way, with seven components in the PCA, we have captured 84.1% of the variance.

4. Factor analysis

For factor analysis we start by trying to do it with all the numerical variables that we had, but even as we changed the number of factors, type of rotation or type of scores, we couldn't get a model that explained a good amount of the variance and had a good fit. We decided to try and remove some of the variables that are not well explained by the model. To do that we need to check the uniqueness. Making factor analysis using different sets of variables, we found two models that can work for us, but they have different purposes.

Best overall model:

```
fa_result_3 <- factanal(final_cleaned_data3, factors = 3, rotation="varimax",
scores="Bartlett", lower = 0.01)
```

Model based on the variables Age, Weight, Height, BMI, BPSysAve, and PulseAgeRatio. We got a model that explains 85.1% of the variance and has the lowest uniqueness, which means that it captures the underlying patterns the best. However, it possesses the risk of overfitting. If we were to not use this model for predicting new data, it is a very good model.

The model's uniqueness:

Age	Weight	Height	BMI	BPSysAve	PulseAgeRatio
0.0100	0.0100	0.0100	0.0100	0.7664	0.0923

This shows that the model explains the variance in the variables Age, Weight, Height, BMI, and PulseAgeRatio very well. BPSysAve is not well explained judging by the high value of uniqueness.

The model's loadings tells us that:

Factor 1 - explains 35.6% variance

Interpretation of F1: Factor 1 captures the variables, which are related to age and health indicators that are affected by age, like BPSysAve and PulseAgeRatio. This factor could represent the health of a person related to their age.

Factor 2 - explains 30.1% variance

Interpretation of F2: Factor 2 appears to represent the body mass or body composition of a person based on the high loadings of the Weight and BMI variables.

Factor 3 - explains 19.4% variance

Interpretation of F3: Factor 3 is probably a "Height" factor that captures the variance in height with minor contribution of weight.

Best model for predicting new data:

```
fa_result_2_f3 <- factanal(final_cleaned_data2, factors = 3, rotation="varimax",
scores="Bartlett", lower = 0.01)
```

Model based on Age, Weight, Height, BMI, BPSysAve, BPDiaAve, and PulseAgeRatio. The best model is a good balance between variance (73.8%), fit, and complexity. The risk of overfitting is less than with the previous model.

The uniqueness of the model:

Age	Weight	Height	BMI	BPSysAve	BPDiaAve	PulseAgeRatio
0.0100	0.0100	0.0100	0.0100	0.766	0.941	0.0922

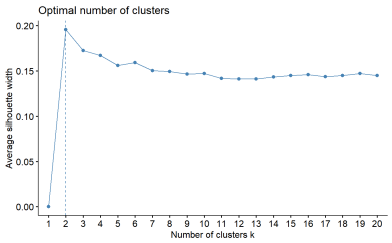
The model explains the variance in the variables Age, Weight, Height, BMI, and PulseAgeRatio very well. BPSysAve and BPDiaAve are not well explained judging by the high value of uniqueness.

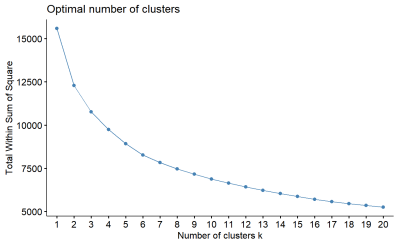
The model's loadings tells us:

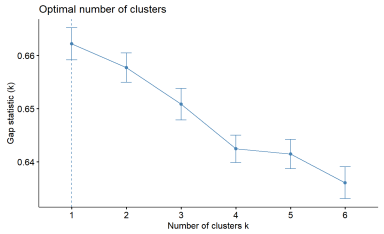
- **Factor 1** is about health related to age. Represents 30.5% variance.
- **Factor 2** represents the body mass or body composition of a person. Represents 26.4% variance
- **Factor 3** is about the height of the person with a small contribution of the weight. Represents 16.9% variance.

5. Clustering

For clustering we wanted to focus on dividing the individuals into groups based on their risk of cardiovascular issues. To do this we selected the variables BMI, BPSysAve, BPDiaAve, TotChol, and Pulse. We decided on using k means for our clustering. Using different approaches, we tried to find the ideal number of clusters.

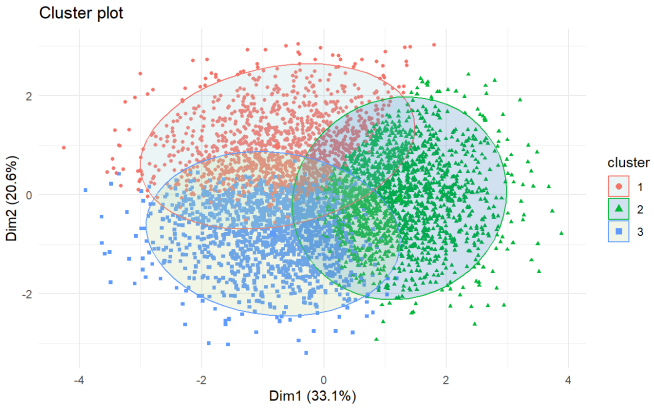






The three graphs give us slightly different answers so we can try and do clustering with 2 centers and with 3 centers and after that decide which is better.

3 clusters



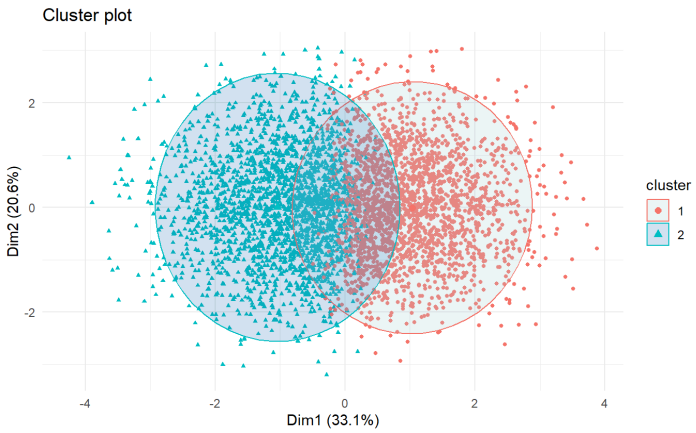
Based on the centers we can interpret what each cluster will represent.

<u>Cluster 1</u> has 1256 individuals. This cluster shows a group that has higher BMI and pulse rate and close to average blood pressure and cholesterol levels. These individuals could have an average risk of cardiovascular issues.

Cluster 2 cotains 1718 people. This cluster shows healthier individuals with below average BMI, blood pressure, cholesterol and pulse. This could indicate that they have a lower risk of cardiovascular issues.
Cluster 3 has 1398 individuals.This cluster shows individuals with normal BMI, but higher blood pressure and cholesterol levels. They might have a higher risk for cardiovascular conditions and could benefit from changing their diet and contacting a doctor to find a way to regulate their blood pressure.
The silhouettes width values are close to zero which indicates that the clusters may overlap.

```
Cluster sizes and average silhouette widths:
     1256      1718      1398
0.1253348 0.2193853 0.1574696
```

2 clusters



Based on the centers we can interpret what each cluster will represent.
Cluster 1 has 2194 individuals. This cluster shows people that are healthier and have below average BMI, blood pressure, cholesterol levels and pulse. They could have lower risk for cardiovascular issues.
Cluster 2 has 2178 individuals. This cluster shows people that have higher than average BMI, blood pressure, cholesterol levels, and pulse. This could indicate a higher chance of cardiovascular issues. They would benefit from changing their lifestyle and contacting a doctor.
The silhouette width values are again close to zero, but they are better than for 3 clusters. From this we can assume that the clustering with only 2 centers is better than the one with 3.

Lastly, we decided to try using different clustering methods and compare them to the k means and see their silhouette widths.

	adjustedRandIndex	Silhouette widths
PAM	0.7916	Worse than k means
Kernel k-means	0.6704373	Worse than k means
EM clustering	0.2492618	Worse than k means

Overall, the k means clustering with 2 clusters seems to be the best classification for our data.